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### Reflection Paper

I saw how the simple properties of an image (such as the dimensions of an image, the data type that each pixel's RGB value is, etc.) relate to the numerical information that the image is composed of. Looking at each of the RGB channels of an image also showed how they could be separated. In addition to looking at each of the RGB channels separately, we also worked with the image in grayscale format, and saw how different mathematical calculations can be performed to alter an image in ways that affect the way the image looks while not altering the image itself in terms of it being a completely different image file.

The lab demonstrated many of the typical image-processing techniques as well as some of the underlying mathematics used to manipulate images. I can see how many typical image-processing techniques as well as many typical image-editing techniques are applied to the pixels of an image. The techniques shown in the lab for adjusting brightness and contrast involved changing the values of the individual pixels. The convolution techniques, including blurring, sharpening, and edge detection showed how kernels can be used to examine the pixels of an image next to each other in order to process them. These two types of processing are very different from the geometric transformations that were also shown in the lab, including translation, scaling, and rotation of the entire image. Histogram processing as well as many different histogram enhancement techniques, including histogram

equalization and CLAHE, also showed how information about an image could be used to manipulate and improve the image.

Additionally, the Nano Banana lab that we had for the set-up of the lab and the first exploration of the computer was very useful in helping me relate to the concepts explored later in the lab. Traditional image-processing uses algorithms to complete specific tasks such as image enhancement, whereas image-generation using Artificial Intelligence (AI) allows images to be generated by learning patterns within images and using that information to generate new images, or even complete tasks such as image editing. In this lab, we explored the traditional way of image-processing and how it relates to the computer vision concepts explored in more detail in future lab sessions.

This lab can be connected to many real world applications including improving or adjusting photographs such as ones taken in low light, medical images or any other type of images. Also edge detection or other image filtering techniques can be used in many different computer vision applications. Geometric transformations can be used for image resizing or to adjust images that have been tilted in order to get them to be correctly aligned. For my future project I would like to see where traditional image processing stands when merged with AI image processing to see where the line is between the two in terms of recognizable images and in terms of how the AI recognizes images.

This lab provided me with an entirely new perspective of how to edit photographs using digital photography. Before taking this lab, I thought of photo editing as adding special effects to images. In this lab, we worked with the fundamental operations of digital image processing. I found that these photo editing effects are math operations that are performed on the pixel data of an image. This lab opened my eyes to the power of programming, in particular computer vision. Computer vision uses

programming, math, and images to help a computer recognize and understand images of objects. I would like to explore this topic further and learn more about how to scale up the simple operations we used in this lab to the very complex algorithms of modern AI image processing systems and how they can be used to recognize and process images of objects.